In [55]:

```python
import pandas as pd
import numpy as np
import seaborn as sea
import matplotlib.pyplot as mtn
```

In [56]:

```python
df = pd.read_csv(r'C:\Users\chira\Desktop\All Files\Python CSVs\Freight Data.csv', encoding = 'unicode_escape')
```

In [57]:

```python
df.head()
```

Out[57]:

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost_USD	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes
3	18746	UK	Japan	2024-02-21 01:00:00	168.50	6060.83	10	4.0	Road	16	Perishable	1232.59	547.86	Yes
4	38414	Australia	South Africa	2026-05-20 13:00:00	255.62	5266.63	4	1.0	Sea	8	Hazardous	1357.43	569.53	Yes

In [58]:

```python
pd.isnull(df).sum()
```

Out[58]:

```
Shipment_ID               0
Origin_Country            0
Destination_Country       0
Shipment_Date             0
Shipment_Weight_kg      944
Shipment_Cost_USD       939
Delay_Days                0
Customer_Rating        2361
Shipment_Mode             0
Handling_Time_Days        0
Package_Type              0
Fuel_Cost_USD           933
Insurance_Cost_USD      932
Customs_Clearance         0
dtype: int64
```

In [59]:

```python
df.fillna(df.median(numeric_only = True), inplace = True)
```

In [60]:

```python
pd.isnull(df).sum()
```

Out[60]:

```
Shipment_ID            0
Origin_Country         0
Destination_Country    0
Shipment_Date          0
Shipment_Weight_kg     0
Shipment_Cost_USD      0
Delay_Days             0
Customer_Rating        0
Shipment_Mode          0
Handling_Time_Days     0
Package_Type           0
Fuel_Cost_USD          0
Insurance_Cost_USD     0
Customs_Clearance      0
dtype: int64
```

In [61]:

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 47000 entries, 0 to 46999
Data columns (total 14 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Shipment_ID          47000 non-null  int64
 1   Origin_Country       47000 non-null  object
 2   Destination_Country  47000 non-null  object
 3   Shipment_Date        47000 non-null  object
 4   Shipment_Weight_kg   47000 non-null  float64
 5   Shipment_Cost_USD    47000 non-null  float64
 6   Delay_Days           47000 non-null  int64
 7   Customer_Rating      47000 non-null  float64
 8   Shipment_Mode        47000 non-null  object
 9   Handling_Time_Days   47000 non-null  int64
 10  Package_Type         47000 non-null  object
 11  Fuel_Cost_USD        47000 non-null  float64
 12  Insurance_Cost_USD   47000 non-null  float64
 13  Customs_Clearance    47000 non-null  object
dtypes: float64(5), int64(3), object(6)
memory usage: 5.0+ MB
```

In [62]:

```python
df['Shipment_Date'] = pd.to_datetime(df['Shipment_Date'], format = 'mixed')
```

In [63]:

```python
df.describe()
```

Out[63]:

	Shipment_ID	Shipment_Date	Shipment_Weight_kg	Shipment_Cost_USD	Delay_Days	Customer_Rating	Handling_Time_Days	Fuel_Cost_USD	Insurance_Cost_USD
count	47000.000000	47000	47000.000000	47000.000000	47000.000000	47000.000000	47000.000000	47000.000000	47000.000000
mean	22502.043383	2024-07-26 13:02:36.178723328	199.755866	5003.887013	6.997787	3.191532	9.970213	1499.301641	500.216348
min	1.000000	2022-01-01 00:00:00	11.140000	249.520000	0.000000	1.000000	1.000000	288.830000	89.450000
25%	11262.750000	2023-04-15 05:45:00	166.600000	4216.037500	3.000000	2.000000	5.000000	1302.937500	433.827500
50%	22504.000000	2024-07-26 15:00:00	199.600000	4997.230000	7.000000	3.000000	10.000000	1498.670000	500.485000
75%	33738.250000	2025-11-06 17:15:00	232.670000	5789.000000	11.000000	4.000000	15.000000	1696.307500	566.560000
max	45000.000000	2027-02-18 23:00:00	393.000000	10227.900000	14.000000	5.000000	19.000000	2740.600000	983.510000
std	12985.652381	NaN	49.703006	1181.478671	4.324730	1.170826	5.475334	296.175316	99.536292

1. Detect and handle outliers in 'Shipment_Cost_USD' using the IQR method.

In [64]:

```python
q1,q3 = np.percentile(df['Shipment_Cost_USD'], (25,75))
```

In [65]:

```python
q1,q3
```

Out[65]:

```
(4216.0375, 5789.0)
```

In [66]:

```python
iqr = q3-q1
```

In [67]:

```python
lower =  q1 - 1.5 *iqr
upper =  q3 + 1.5 *iqr
```

In [68]:

```python
lower, upper
```

Out[68]:

```
(1856.593750000001, 8148.443749999999)
```

In [69]:

```python
df = df[(df['Shipment_Cost_USD'] <= upper) & (df['Shipment_Cost_USD'] >= lower)]
```

In [70]:

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 46635 entries, 0 to 46999
Data columns (total 14 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Shipment_ID          46635 non-null  int64
 1   Origin_Country       46635 non-null  object
 2   Destination_Country  46635 non-null  object
 3   Shipment_Date        46635 non-null  datetime64[ns]
 4   Shipment_Weight_kg   46635 non-null  float64
 5   Shipment_Cost_USD    46635 non-null  float64
 6   Delay_Days           46635 non-null  int64
 7   Customer_Rating      46635 non-null  float64
 8   Shipment_Mode        46635 non-null  object
 9   Handling_Time_Days   46635 non-null  int64
 10  Package_Type         46635 non-null  object
 11  Fuel_Cost_USD        46635 non-null  float64
 12  Insurance_Cost_USD   46635 non-null  float64
 13  Customs_Clearance    46635 non-null  object
dtypes: datetime64[ns](1), float64(5), int64(3), object(5)
memory usage: 5.3+ MB
```

2. What is the overall distribution of shipment costs?

In [71]:

```python
df.head(2)
```

Out[71]:

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost_USD	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes

In [72]:

```python
df.rename(columns = {'Shipment_Cost_USD' : 'Shipment_Cost'}, inplace = True)
```

In [73]:

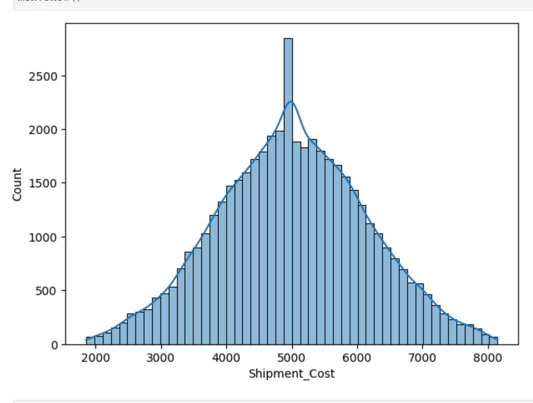
```python
df.head(2)
```

Out[73]:

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes

In [74]:

```python
mtn.figure(figsize = (7,5))
sea.histplot(df['Shipment_Cost'], bins = 50, kde = True)
mtn.show()
```



In [77]:

```python
mtn.figure(figsize = (6,5))
sea.boxplot(y = 'Shipment_Cost', data = df, color = 'brown')
mtn.title('Distribution of Shipment Cost')
mtn.show()
```

3. What are the top 5 countries with the highest average shipment cost?

In [78]:

```python
df.head(3)
```

Out[78]:

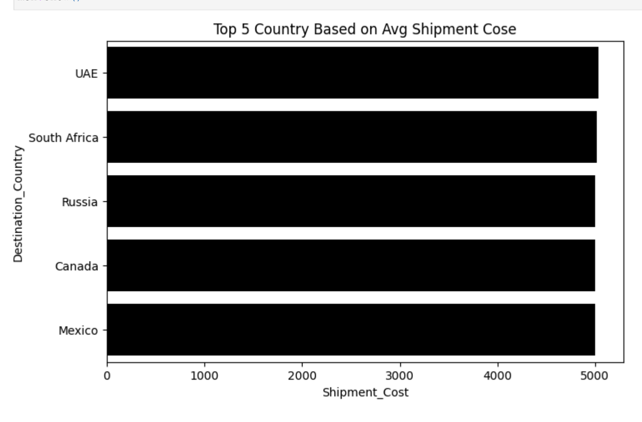
	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes

In [86]:

```python
Top5 = round(df.groupby('Destination_Country')['Shipment_Cost'].mean().reset_index().sort_values(by = 'Shipment_Cost', ascending = False),1).head()
```

In [91]:

```python
mtn.figure(figsize = (8,5))
sea.barplot(y = 'Destination_Country', x = 'Shipment_Cost', data = Top5, color = 'black')
mtn.title('Top 5 Country Based on Avg Shipment Cose')
mtn.show()
```



4. How does shipment cost vary across different shipment modes?

In [92]:

```python
df.head(2)
```

Out[92]:

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes

In [94]:

```python
Modes = df.groupby('Shipment_Mode')['Shipment_Cost'].sum().reset_index()
```

In [100…

```python
Modes['Shipment_Cost in M'] = round(Modes['Shipment_Cost']/1000000,1)
```

In [101…

```python
Modes
```

Out[101…

	Shipment_Mode	Shipment_Cost	Shipment_Cost in M
0	Air	58505268.78	58.5
1	Rail	58676773.49	58.7
2	Road	58149597.82	58.1
3	Sea	58107099.59	58.1

5. Which package type has the highest delay on average?

In [102…

```python
df.head(2)
```

Out[102…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes

In [103…

```python
df.groupby('Package_Type')['Delay_Days'].mean().reset_index().sort_values(by = 'Delay_Days', ascending = False)
```

Out[103…

	Package_Type	Delay_Days
1	Hazardous	7.020191
2	Perishable	7.020178
3	Standard	6.984438
0	Fragile	6.971232

6. What is the correlation between shipment weight and shipment cost?

In [104…

```python
df.head(2)
```

Out[104…

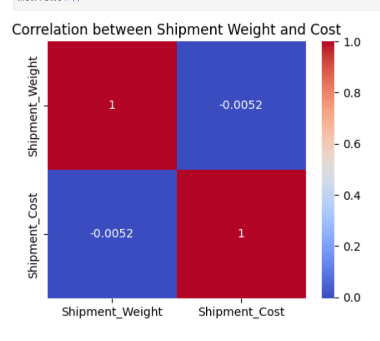
	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight_kg	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes

In [105…

```python
df.rename(columns = {'Shipment_Weight_kg' : 'Shipment_Weight'}, inplace = True)
```

In [107…

```python
mtn.figure(figsize = (5,4))
sea.heatmap(df[['Shipment_Weight', 'Shipment_Cost']].corr(), annot = True, cmap = 'coolwarm')
mtn.title('Correlation between Shipment Weight and Cost')
mtn.show()
```



7. Analyze the trend of shipments over time (monthly/yearly basis).

In [108…

```python
df.head(5)
```

Out[108…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes
3	18746	UK	Japan	2024-02-21 01:00:00	168.50	6060.83	10	4.0	Road	16	Perishable	1232.59	547.86	Yes
4	38414	Australia	South Africa	2026-05-20 13:00:00	255.62	5266.63	4	1.0	Sea	8	Hazardous	1357.43	569.53	Yes

In [109…

```python
df['Months'] = df['Shipment_Date'].dt.month_name()
```

In [113…

```python
df['Year'] = df["Shipment_Date"].dt.year
```

In [117…

```python
df['month_num'] = df['Shipment_Date'].dt.month
```

In [120…

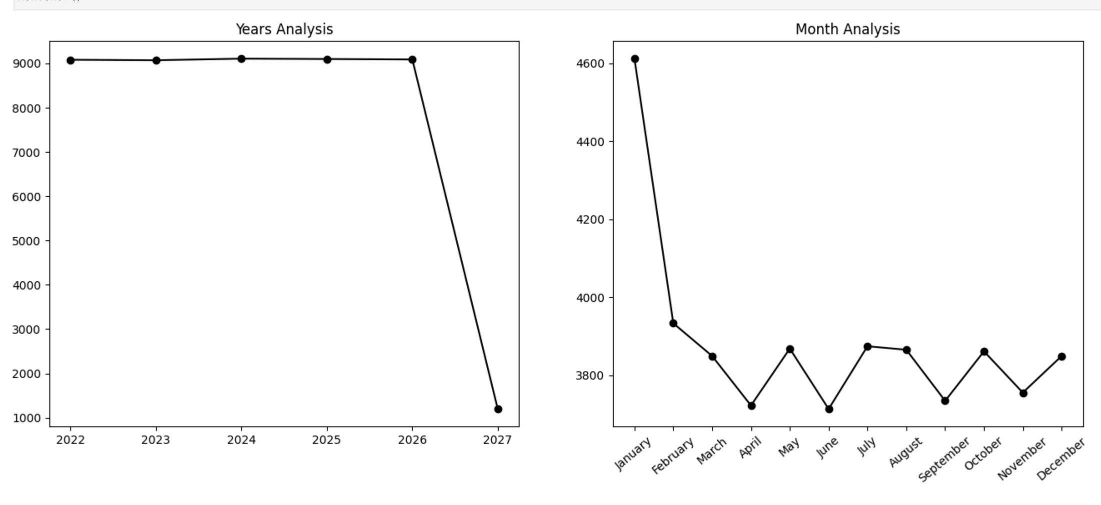
```python
Years = df.groupby('Year').size().reset_index(name = 'Shipments')
```

In [119…

```python
months = df.groupby(['Months', 'month_num']).size().reset_index(name = 'Shipments').sort_values(by = 'month_num', ascending = True)
```

In [125…

```python
mtn.figure(figsize = (16,6))
mtn.subplot(1,2,1)
mtn.plot(Years['Year'], Years['Shipments'], color = 'black', marker = 'o')
mtn.title('Years Analysis')
mtn.subplot(1,2,2)
mtn.plot(months['Months'], months['Shipments'], color = 'black', marker = 'o')
mtn.xticks(rotation = 40)
mtn.title('Month Analysis')
mtn.show()
```



8. What percentage of shipments require customs clearance?

In [127…

```python
df.head(2)
```

Out[127…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8

In [134…

```python
round(df['Customs_Clearance'].value_counts(normalize = True)* 100,1)
```

Out[134…

```
Customs_Clearance
Yes    85.0
No     15.0
Name: proportion, dtype: float64
```

9. Identify the busiest shipment routes (Origin-Destination combinations).

In [135…

```python
df.head(3)
```

Out[135…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes	July	2026	7

In [140…

```python
df['Shipment Routes'] = df['Origin_Country'] +  ' - ' + df["Destination_Country"]
```

In [141…

```python
df.head(3)
```

Out[141…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num	Shipment Routes
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6	Germany - Russia
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8	USA - Canada
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes	July	2026	7	Australia - Russia

In [142…

```python
df.groupby('Shipment Routes').size().reset_index(name = 'Total Shipments').sort_values(by = 'Total Shipments', ascending = False).head()
```

Out[142…

	Shipment Routes	Total Shipments
13	Brazil - UAE	1022
7	Brazil - Canada	1016
39	UK - Russia	1013
15	China - France	996
36	UK - France	994

Brazil and UAE are the most busiest shipment routes

10. Find the top 3 most cost-efficient countries (lowest average shipment cost).

In [143…

```python
df.head(4)
```

Out[143…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num	Shipment Routes
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6	Germany - Russia
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8	USA - Canada
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes	July	2026	7	Australia - Russia
3	18746	UK	Japan	2024-02-21 01:00:00	168.50	6060.83	10	4.0	Road	16	Perishable	1232.59	547.86	Yes	February	2024	2	UK - Japan

In [144…

```python
df.groupby('Origin_Country')['Shipment_Cost'].mean().reset_index().sort_values(by = 'Shipment_Cost', ascending = True)
```

Out[144…

	Origin_Country	Shipment_Cost
3	Germany	4988.906367
5	UK	4998.872779
6	USA	5001.656161
4	India	5004.785408
0	Australia	5007.934007
2	China	5010.607682
1	Brazil	5026.129168

11. Predict missing 'Customer_Rating' values using the median rating per country.

In [145…

```python
df.head(2)
```

Out[145…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num	Shipment Routes
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6	Germany - Russia
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8	USA - Canada

In [146…

```python
df.groupby('Destination_Country')['Customer_Rating'].median().reset_index()
```

Out[146…

	Destination_Country	Customer_Rating
0	Canada	3.0
1	France	3.0
2	Japan	3.0
3	Mexico	3.0
4	Russia	3.0
5	South Africa	3.0
6	UAE	3.0

12. Does shipment weight significantly impact shipment cost? Perform hypothesis testing.

In [148…

```python
df.head(3)
```

Out[148…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num	Shipment Routes
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6	Germany - Russia
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8	USA - Canada
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes	July	2026	7	Australia - Russia

In [149…

```python
from scipy.stats import pearsonr
```

In [150…

```python
corr, p_value = pearsonr(df['Shipment_Cost'], df['Shipment_Weight'])
```

In [151…

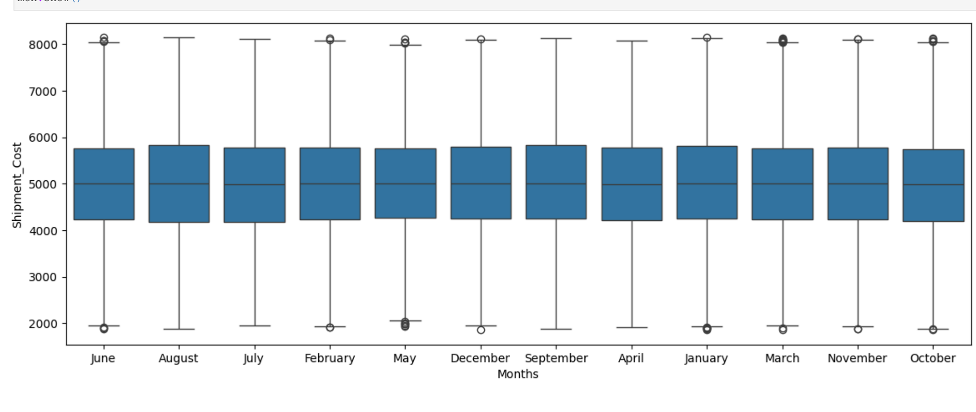
```python
print(f"correlation : {corr}, P_Value : {p_value}")
```

```
correlation : -0.005181638594122708, P_Value : 0.2631574001950072
```

13. Analyze the seasonal impact on shipment volume.

In [156…

```python
mtn.figure(figsize = (14,5))
sea.boxplot(x='Months', y='Shipment_Cost', data=df)
mtn.show()
```



14. Can we identify anomalies in shipment costs that could indicate fraud, pricing errors, or unexpected surges?

In [159…

```python
df.head(3)
```

Out[159…

	Shipment_ID	Origin_Country	Destination_Country	Shipment_Date	Shipment_Weight	Shipment_Cost	Delay_Days	Customer_Rating	Shipment_Mode	Handling_Time_Days	Package_Type	Fuel_Cost_USD	Insurance_Cost_USD	Customs_Clearance	Months	Year	month_num	Shipment Routes
0	30565	Germany	Russia	2025-06-27 12:00:00	181.77	4809.36	5	4.0	Air	6	Standard	1777.50	568.38	Yes	June	2025	6	Germany - Russia
1	23352	USA	Canada	2024-08-30 23:00:00	185.75	4151.59	11	3.0	Rail	6	Hazardous	1779.82	450.07	Yes	August	2024	8	USA - Canada
2	39974	Australia	Russia	2026-07-24 13:00:00	223.32	4318.30	9	5.0	Sea	1	Standard	1639.28	562.50	Yes	July	2026	7	Australia - Russia

In [157…

```python
from sklearn.ensemble import IsolationForest
```

In [160…

```python
features = df[['Shipment_Cost', 'Shipment_Weight', 'Handling_Time_Days']].dropna()
```

In [162…

```python
iso_forest = IsolationForest(contamination = 0.02, random_state = 42)
features['Anomaly'] = iso_forest.fit_predict(features)
```

In [163…

```python
anomalies = features[features['Anomaly'] == -1]
```

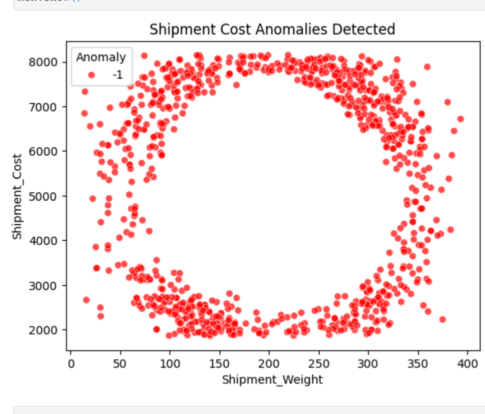
In [164…

```python
print("Detected Anomalies in Shipment Costs:")
print(anomalies.head())
```

```
Detected Anomalies in Shipment Costs:
     Shipment_Cost  Shipment_Weight  Handling_Time_Days  Anomaly
30         7754.64           297.22                  16       -1
39         4234.31           382.95                   5       -1
89         3509.94           358.76                  19       -1
102        4142.36           373.03                   6       -1
182        7226.28           265.43                  19       -1
```

In [168…

```python
# Visualizing anomalies
sea.scatterplot(x='Shipment_Weight', y='Shipment_Cost', hue= 'Anomaly', data = anomalies,
                palette={1: 'blue', -1: 'red'}, alpha=0.7)
mtn.title("Shipment Cost Anomalies Detected")
mtn.show()
```



In [ ]: